
Exercises in Theoretical and Empirical Microeconomics and Macroeconomics Using State-of-the-Art Econometric Methods

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Outline

1. Introduction

2. Boxes (block environments) and columns

1. Boxes (block environments)
2. Using columns and blocks for comparisons

3. Data

4. Testing various elements

5. Color palettes of various universities from several continents

6. Results

7. Conclusion

8. Math “Torture” Test

9. References

1

Introduction

Introduction

- Theory predicts: Something very specific (Sarfraz and Razzak 2002).
- In practice,
 - we observe this;
 - we also observe that.
- This underlines the importance of analyzing **left-digit bias** (Aristotle 1968).
- *Serif* font *Sans-serif* font *Monospaced* font
- ***Serif* font *Sans-serif* font *Monospaced* font**

Introduction: Our contribution

1. This is a nested numbered list.
 - a) Describe left-digit bias in a different setting: A.
 - b) Describe left-digit bias in a different setting: B.
 - i. Describe left-digit bias in setting B.1.
 - ii. Describe left-digit bias in setting B.2.
 - iii. Describe left-digit bias in setting B.3.
 - iv. Describe left-digit bias in setting B.4.
2. This is another bullet point.
3. And another bullet point.

Inserting `\par` before the beginning of a lower-level list is necessary for consistent spacing, see <https://stackoverflow.com/questions/75483326/latex-beamer-nested-itemize-vertical-spacing-isnt-constant>.

Questions to Structure Your Presentation (1/3)

You may find the following list of questions helpful as a guide to making sure that your presentation covers all important aspects. The questions equally apply, of course, to posters and manuscripts.

Questions to Structure Your Presentation (2/3)

Your introduction might answer the following questions (not necessarily in this order, but this order will produce a “funnel” structure):

1. What do we already know?
2. What do we not know yet (the “knowledge gap”)?
3. Why is it interesting/relevant to close the gap? [“No one has done it yet” ist not enough.]
- 4. What is the research question? [R]**
5. In which way(s) does answering the research question contribute to closing the knowledge gap?
6. Which method(s) do you use to answer the research question?
7. Why is/are the chosen method(s) (more) suitable (than potential alternative methods)?

Questions to Structure Your Presentation (3/3)

Your conclusion/discussion might answer the following questions (the following order will produce a “reverse funnel” structure):

8. What was the knowledge gap before the current study? [Similar to the introduction.]
9. **What did you find, and how does it contribute to closing the knowledge gap? [A]**
Which method/data did you use; what are the results? [Recap & take-home message.]
10. How do your results relate to (confirm/contradict/complement/extend) the results from previous and other contemporaneous studies?
11. What part of the gap is still open?
Phrased a bit more negatively: What are the limitations of your approach?
12. Next steps/avenues for future research: How could we go about closing the remaining knowledge gap (by removing the limitations of the approaches used so far)?

John Cochrane might do away with question 12. See his “Writing Tips for Ph.D. Students,” https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf.

More Structuring Advice

In her book *The Little Book of Research Writing*, Varanya Chaubey proposes a simpler list of only three items—the **"RAP method"**:

- The **"R"** stands for "research question."
- The **"A"** stands for "answer" (to the research question, of course).
- The **"P"** stands for "positioning statement." (How does our manuscript relate to the literature: Does it corroborate or extend or contradict others' findings?)

Relating the "RAP" approach to the the list of questions above, question 4 is the "R," the answer to question 9 is the "A," and the remaining questions spell out the "P."

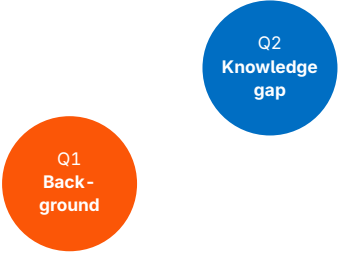
For a quick summary of the book, see

https://mauve-porcupine-8992.squarespace.com/s/Chaubey_Research_Writing-bxddd.pdf.

See also <https://www.econscribe.org/about> and <https://arxiv.org/pdf/2012.07787>.

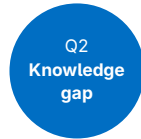
An orange circle with a solid fill, containing the text "Q1 Back-ground" in white. The text is centered within the circle.

Q1
Back-
ground



Q2
Knowledge
gap

Q1
Back-
ground





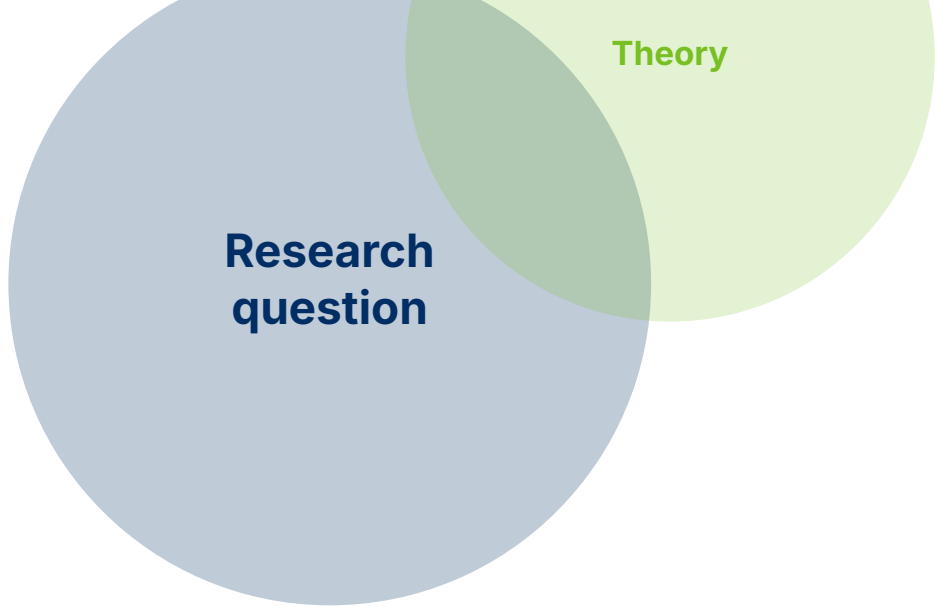








Research question

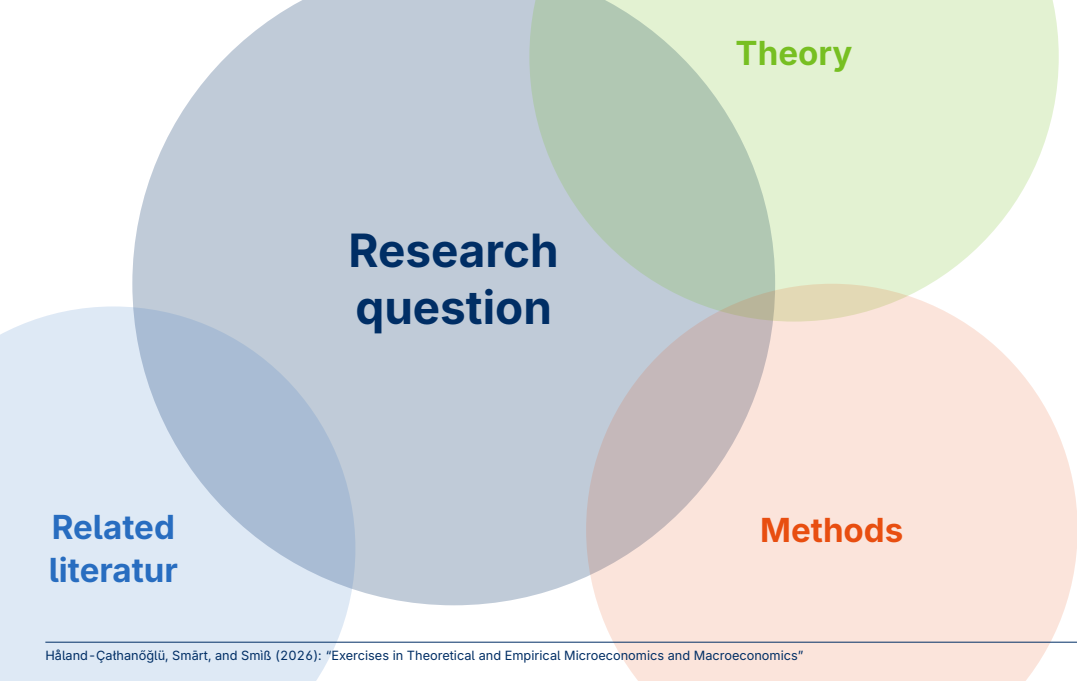


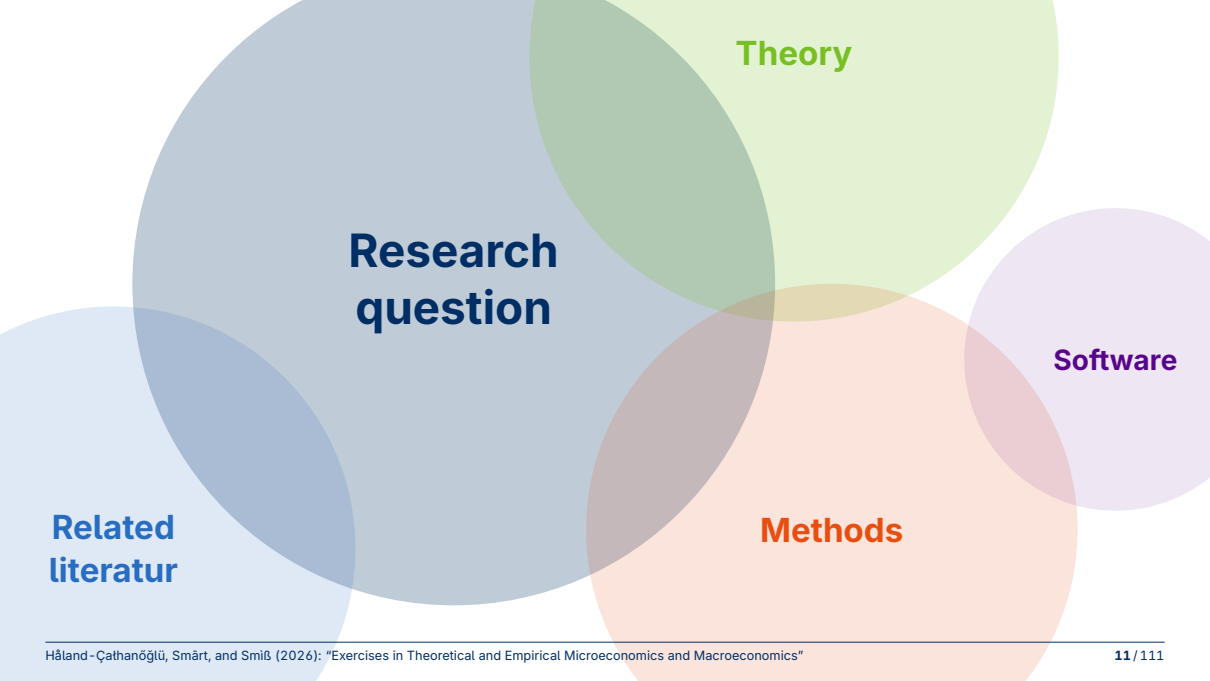


**Research
question**

Theory

**Related
literatur**





**Research
question**

Theory

Software

Methods

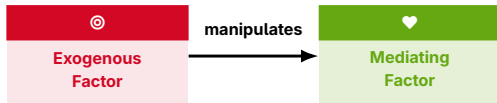
**Related
literatur**

Another example TikZ graphic



**Exogenous
Factor**

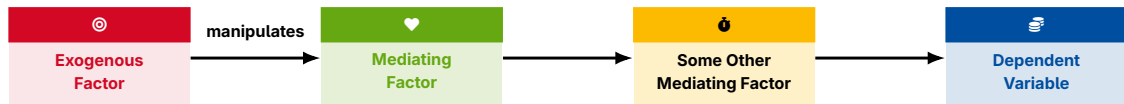
Another example TikZ graphic



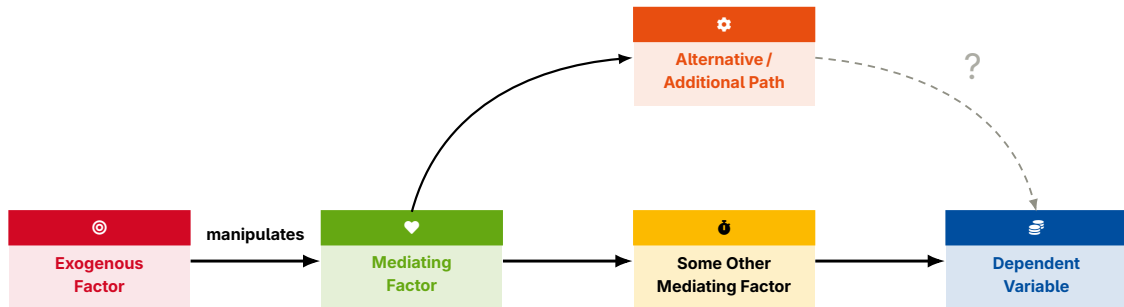
Another example TikZ graphic



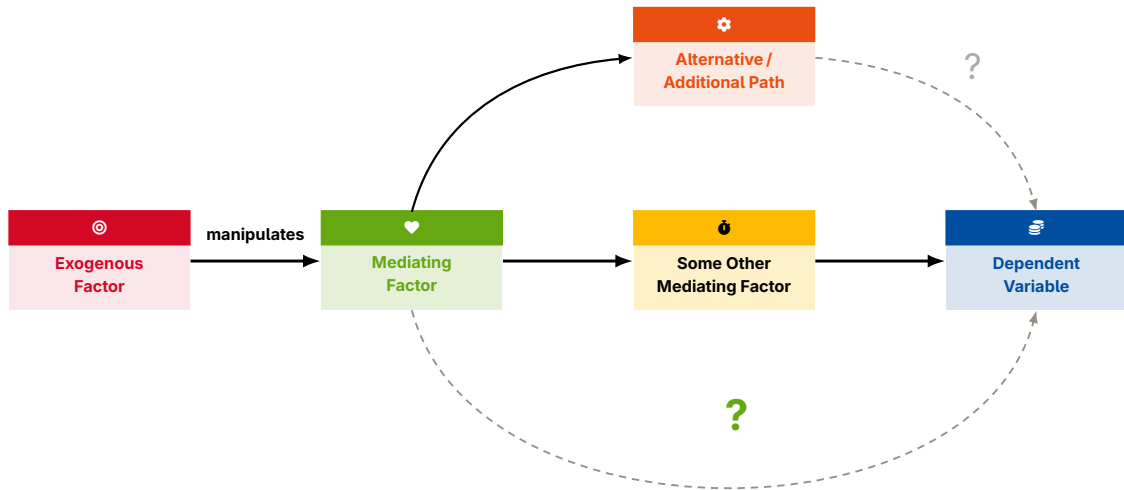
Another example TikZ graphic



Another example TikZ graphic



Another example TikZ graphic



Introduction: Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Introduction: Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Category B

- van Gennep (1909)

Introduction: Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Category B

- van Gennep (1909)

Category C

- Gerhardt (2000), Chiu and Chow (1978)

Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Category B

- Van Gennep (1909)

Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Category B

- Van Gennep (1909)

Category C

- Gerhardt (2000), Chiu and Chow (1978)

No list

Item 1 as plain body text.

Item 2 as plain body text.

With the beamer default settings, the text is positioned too close to the top, and the vertical space between paragraphs (items) is too small (namely, absent).

Changing `\parskip` could fix this but has undesirable consequences in other places. For this reason, one should place *any* text in a `list` environment.

list environment, followed by other environments

Item 1 in list environment.

Item 2 in list environment.

Item 3 in list environment.

Item 1 in subsequent list environment.

Item 2 in subsequent list environment.

- Item 1 in subsequent itemize environment.
- Item 2 in subsequent itemize environment.

1. Item 1 in subsequent enumerate environment.

2. Item 2 in subsequent enumerate environment.

- Item 1 in subsequent itemize environment.
- Item 2 in subsequent itemize environment.

itemize environment, followed by other environments

- Item 1 in itemize environment.
- Item 2 in itemize environment.

Item 1 in subsequent list environment.

Item 2 in subsequent list environment.

Item 3 in subsequent list environment.

Item 1 in subsequent list environment.

Item 2 in subsequent list environment.

1. Item 1 in subsequent enumerate environment.

2. Item 2 in subsequent enumerate environment.

Item 1 in subsequent list environment.

Item 2 in subsequent list environment.

itemize environment

- Item 1.
 - Item 1.
 - Item 2.
- Item 2.

Item 1 in subsequent list environment.

Item 2 in subsequent list environment.

enumerate environment

1. Item 1.

a) Item 1.

b) Item 2.

2. Item 2.

Body text.

Also here: vertical space between paragraphs is absent. Hence, better to use `list` environment.

enumerate (1/2)

1. Item 1.

a) Item 1.

b) Item 2.

enumerate (2/2)

2. Item 2.

a) Item 1.

b) Item 2.

Item 1 in subsequent list environment.

Item 2 in subsequent list environment.

Testing \framebreaks (1/2)

1. Before frame break.

Testing `\framebreak` (2/2)

2. After frame break—vertical spacing should be consistent.

Testing \framebreaks (1/2)

Before frame break.

Testing `\framebreaks` (2/2)

After frame break—vertical spacing should be consistent.

Testing \framebreaks (1/3)

- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
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- Vertical spacing should be consistent.

Testing \framebreaks (2/3)

- Vertical spacing should be consistent.
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- Vertical spacing should be consistent.
- Vertical spacing should be consistent.

Testing `\framebreaks` (3/3)

- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.

An automated animation

The automated transition to the next slide (= page in the PDF document) only works in full-screen mode.

- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, July 18, 2026) not available in macOS Preview, Skim, and SumatraPDF.



Figure 1. Step 1—Angle: 30.0°

► Jump to the end

An automated animation

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Figure 1. Step 2—Angle: 60.0°

An automated animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, July 18, 2026) not available in macOS Preview, Skim, and SumatraPDF.



Figure 1. Step 3—Angle: 90.0°

An automated animation

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Figure 1. Step 4—Angle: 120.0°

An automated animation

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Figure 1. Step 5—Angle: 150.0°

An automated animation

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Figure 1. Step 6—Angle: 180.0°

An automated animation

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- Unfortunately, it is (currently, July 18, 2026) not available in macOS Preview, Skim, and SumatraPDF.



Figure 1. Step 7—Angle: 210.0°

An automated animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, July 18, 2026) not available in macOS Preview, Skim, and SumatraPDF.



Figure 1. Step 8—Angle: 240.0°

An automated animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, July 18, 2026) not available in macOS Preview, Skim, and SumatraPDF.



Figure 1. Step 9—Angle: 270.0°

An automated animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, July 18, 2026) not available in macOS Preview, Skim, and SumatraPDF.



Figure 1. Step 10—Angle: 300.0°

An automated animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, July 18, 2026) not available in macOS Preview, Skim, and SumatraPDF.



Figure 1. Step 11—Angle: 330.0°

An automated animation

The automated transition to the next slide (= page in the PDF document) only works in full-screen mode.

- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, July 18, 2026) not available in macOS Preview, Skim, and SumatraPDF.

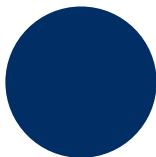


Figure 1. Step 12—Angle: 360.0°

◀ Back to the start

2

**Boxes (block environments)
and columns**

blocks (1)

This is a standard block

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

This is an alertblock

Use it for information that you want to stand out even more.

1. Use it for information that you want to stand out even more.
2. Use it for information that you want to stand out even more.

Theorem (Pythagorean theorem)

In any right triangle, the area of the square whose side is the hypotenuse is equal to the sum of the areas of the squares on the other two sides. Algebraically, $x^2 + y^2 = z^2$.

Definition (Pythagorean triple)

A Pythagorean triple represents the lengths of the sides of a right triangle where all three sides have integer lengths. Such a triple is commonly written (a, b, c) .

Examples of Pythagorean triples

Some well-known examples of Pythagorean triples are $(3, 4, 5)$ and $(5, 12, 13)$.

Lemma

Given two line segments whose lengths are a and b , respectively, there is a real number r such that $b = r a$.

Proof.

To prove a lemma by contradiction, try and assume that the statement is false, proceed from there and at some point you will arrive at a contradiction. ■

Using columns and blocks for comparisons

This is a standard block

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

This is an alertblock

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

This is an exampleblock

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

Using columns and blocks for comparisons

This is a standard block

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

This is an alertblock

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

This is an exampleblock

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

Insurance: What is traded?

We are very used to designing slides linearly from top to bottom.

For instance, we often put (contrasting) juxtapositions in lists—like this:

- **Obvious answer:** a certain payment, the premium, is exchanged for the promise to pay partial or full compensation (cover) for loss resulting from carefully specified events. So premium is exchanged for cover.
- **Less obvious answer:** state - contingent incomes.
Income is reduced in states of the world in which the specified events do not happen, and increased in states in which the events do happen, as compared to the situation without insurance.

Insurance: What is traded?

It often makes more sense, however, to design (contrasting) juxtapositions by putting the elements side by side—like this:

Obvious answer

A certain payment, the premium, is exchanged for the promise to pay partial or full compensation (cover) for loss resulting from carefully specified events.

So premium is exchanged for cover.

Less obvious answer

State-contingent incomes.

Income is reduced in states of the world in which the specified events do not happen, and increased in states in which the events do happen, as compared to the situation without insurance.

3

Data

- Goal: Identify left-digit bias.
- Bla bla.
- We follow the approach by Goossens et al. (1994):
 - criterion 1,
 - criterion 2,
 - criterion 3.
- Bla bla.
- Bla bla.

Summary statistics

	L-D	D-B	B-L	D-L
Panel A: Description A				
Characteristic 1	46.3 (10.7)	47.6 (10.3)	44.2 (11.5)	43.5 (12.0)
Characteristic 2	97.2 (48.6)	109.9 (90.1)	90.0 (69.2)	78.8 (32.5)
Characteristic 3	7.8 (6.4)	8.7 (6.8)	7.5 (6.2)	7.7 (6.2)
Characteristic 4	16.3 (8.4)	16.9 (8.2)	15.6 (8.0)	14.7 (8.0)
Characteristic 5	11.9 (1.3)	12.3 (1.8)	12.0 (1.3)	11.9 (1.3)
Panel B: Description B				
Characteristic 1	35.0 (9.9)	32.8 (10.0)	35.0 (9.8)	32.8 (10.3)
Characteristic 2	86.5 (28.0)	79.8 (31.3)	91.5 (33.1)	70.6 (25.9)
Characteristic 4	11.4 (7.1)	10.1 (6.6)	11.8 (7.7)	10.2 (6.9)
Characteristic 5	12.0 (1.4)	12.3 (1.9)	12.5 (1.9)	12.0 (1.4)
Number of observations	13,978	1,477	531	1,675

Summary statistics

	L-D	D-B	B-L	D-L
Panel A: Description A				
Characteristic 1	46.3 (10.7)	47.6 (10.3)	44.2 (11.5)	43.5 (12.0)
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Characteristic 4	16.3 (8.4)	16.9 (8.2)	15.6 (8.0)	14.7 (8.0)
Characteristic 5	11.9 (1.3)	12.3 (1.8)	12.0 (1.3)	11.9 (1.3)
Panel B: Description B				
Characteristic 1	35.0 (9.9)	32.8 (10.0)	35.0 (9.8)	32.8 (10.3)
Characteristic 2	86.5 (28.0)	79.8 (31.3)	91.5 (33.1)	70.6 (25.9)
Characteristic 4	11.4 (7.1)	10.1 (6.6)	11.8 (7.7)	10.2 (6.9)
Characteristic 5	12.0 (1.4)	12.3 (1.9)	12.5 (1.9)	12.0 (1.4)
Number of observations	13,978	1,477	531	1,675

A `tabularray`-based table with animated cell highlighting

	Column 2	Column 3	Column 4	Column 5	Column 6
Row 2	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A `tabularray`-based table with animated cell highlighting

	Column 2	Column 3	Column 4	Column 5	Column 6
Row 2	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A `tabularray`-based table with animated cell highlighting

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Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

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Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A `tabularray`-based table with animated cell highlighting

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Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A `tabularray`-based table with animated cell highlighting

	Column 2	Column 3	Column 4	Column 5	Column 6
Row 2	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3					
Row 4					
Row 5					
Row 6					
Row 7					
Row 8					

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)				
Row 4	(4, 2)				
Row 5	(5, 2)				
Row 6	(6, 2)				
Row 7	(7, 2)				
Row 8	(8, 2)				

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)			
Row 4	(4, 2)	(4, 3)			
Row 5	(5, 2)	(5, 3)			
Row 6	(6, 2)	(6, 3)			
Row 7	(7, 2)	(7, 3)			
Row 8	(8, 2)	(8, 3)			

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)		
Row 4	(4, 2)	(4, 3)	(4, 4)		
Row 5	(5, 2)	(5, 3)	(5, 4)		
Row 6	(6, 2)	(6, 3)	(6, 4)		
Row 7	(7, 2)	(7, 3)	(7, 4)		
Row 8	(8, 2)	(8, 3)	(8, 4)		

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)		
Row 4	(4, 2)	(4, 3)	(4, 4)		
Row 5	(5, 2)	(5, 3)	(5, 4)		
Row 6	(6, 2)	(6, 3)	(6, 4)		
Row 7	(7, 2)	(7, 3)	(7, 4)		
Row 8	(8, 2)	(8, 3)	(8, 4)		

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	(7, 6)
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	(7, 6)
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	(8, 6)

4

Testing various elements

A frame title that is really, really, really, really, really, really, really, really long—works ... but strongly discouraged

A frame subtitle—it is also really, really, really, really, really, really, really, really long. And it works, too. It should still be avoided.

- The coverage chosen is always at a point of tangency between an indifference curve and a budget line, for $q^* > 0$. Thus the cases of **full**, **partial** and **more than full cover** correspond to the cases in which the budget constraint defined by p is respectively **coincident** with, **flatter** than, or **steeper** than the expected value line.
- **Note:** If the budget line is so flat that it does not intersect the indifference curve passing through the initial endowment point, then we have $q^* = 0$.
- **Note:** If only full or zero cover are available, the buyer takes full cover if and only if the resulting point on the certainty line is above the *certainty equivalent* of the initial endowment point at $\tilde{y}$.

Simplest possible model: 2 states (1, 2)

Basic setup for both models | Basic setup for both models | Basic setup for both models

- Income in the no-loss state: $y_1 = y$.
- Income in the loss state: $y_2 = y - L$.
- Probability of loss L is π .
- Expected value of income without insurance:

$$\bar{y} = (1 - \pi) y + \pi (y - L) = y - \pi L$$

- Expected value of income loss: πL
- Expected utility in the absence of insurance:

$$EU = (1 - \pi) u(y) + \pi u(y - L)$$

- In the absence of insurance, the individual has an uncertain income endowment.

Methodology: Assumptions on the utility function

- We assume that the preferences over uncertain alternatives are captured by von Neumann–Morgenstern expected utility over income y :

$$EU = \sum_{i=1}^n \pi_i u(y - L_i) \quad EU = \sum_{i=1}^n \pi_i u(y - L_i)$$

or

$$EU = \int_0^{L_{\max}} f(L) u(y - L_i) dL \quad EU = \int_0^{L_{\max}} f(L) u(y - L_i) dL$$

for the continuous case.

- u_i is uniquely determined up to positive linear transformation.
- u_i is assumed to be three times differentiable and strictly concave in income y : $u'_i(y) > 0$ and $u''_i(y) < 0$.
- Usually, we assume state independence of utility.

Methodology: Assumptions on the utility function

- We assume that the preferences over uncertain alternatives are captured by von Neumann–Morgenstern expected utility over income y :

$$EU = \sum_{i=1}^n \pi_i u(y - L_i) \quad \textcolor{red}{EU} = \sum_{i=1}^n \textcolor{red}{\pi_i} \textcolor{red}{u(y - L_i)}$$

or

$$EU = \int_0^{L_{\max}} f(L) u(y - L_i) dL \quad \textcolor{red}{EU} = \int_0^{\textcolor{red}{L_{\max}}} \textcolor{red}{f(L)} \textcolor{red}{u(y - L_i)} dL$$

for the continuous case.

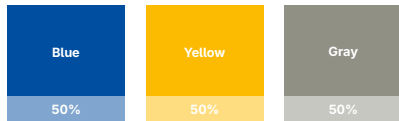
- u_i is uniquely determined up to positive linear transformation
- u_i is assumed to be three times differentiable and strictly concave in income y :
 $u'_i(y) > 0$ and $u''_i(y) < 0$.
- Usually, we assume state independence of utility.

5

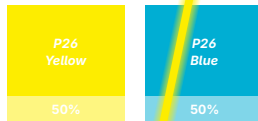
**Color palettes of various
universities from several continents**

University of Bonn Color Palette

Colors

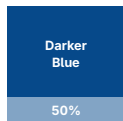


P26 Colors



See https://www.uni-bonn.de/de/universitaet/medien-universitaet/medien-presse-kommunikation/medien-corporate-design/ubo-manual_extern_6_2022.pdf.

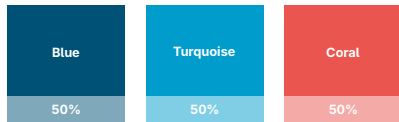
Darker blue (used on uni-bonn.de)



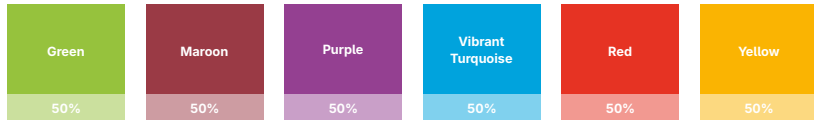
Attention!

University of Cologne Color Palette

Universal colors



Departments' colors

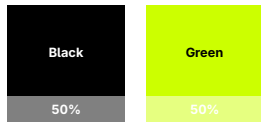


See https://kommunikation-marketing.uni-koeln.de/e135228/e48647/e49192/e49987/Branding_handbook_eng_ger.pdf.

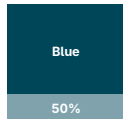
Attention!

Freie Universität Berlin Color Palette

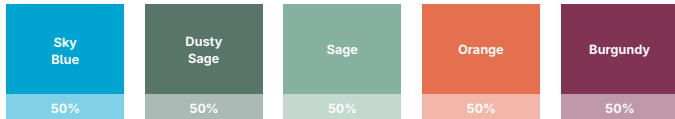
Primary colors



Secondary color



Additional colors

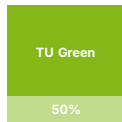


See <https://www.fu-berlin.de/en/sites/corporate-design/guidelines/index.html>.

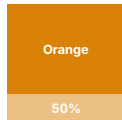
Attention! **Attention!**

TU Dortmund University Color Palette

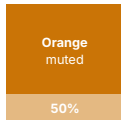
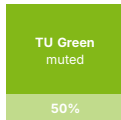
Primary color



Accent color



Web colors

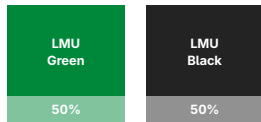


See <https://itmc.tu-dortmund.de/unsere-services/medien-services/grafik/design/corporate-design/>.

Attention!

Ludwig - Maximilians - Universität München Color Palette

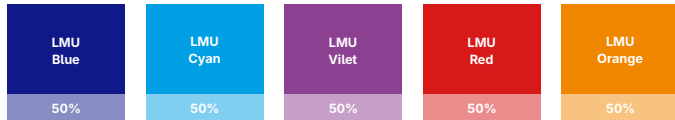
Primary colors



Secondary colors



Accent colors (lmu.de)

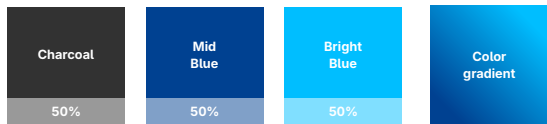


See <https://www.lmu.de/de/die-lmu/struktur/zentrale-universitaetsverwaltung/presse-und-kommunikation-puk/lmu-brand-guide/designgrundsaeetze/farben/>.

Attention!

University of Stuttgart Color Palette

Colors

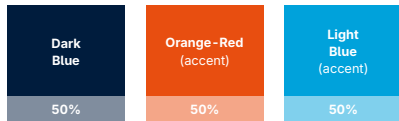


See https://www.beschaefigte.uni-stuttgart.de/uni-services/oeffentlichkeitsarbeit/corporate-design/cd-dateien/Uni_Stuttgart_CD-Manual.pdf.

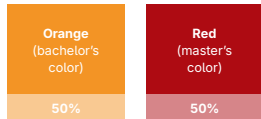
Attention!

Maastricht University Color Palette

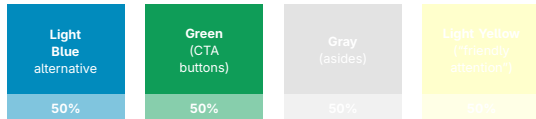
Primary color and accent colors



Bachelor's and master's colors



Colors for specific functions



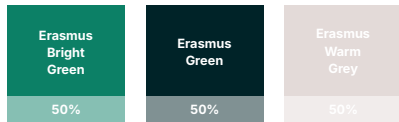
See

<https://www.maastrichtuniversity.nl/support/communications-guide/house-style/colours>.

Attention!

Erasmus University Color Palette (1)

Corporate colors

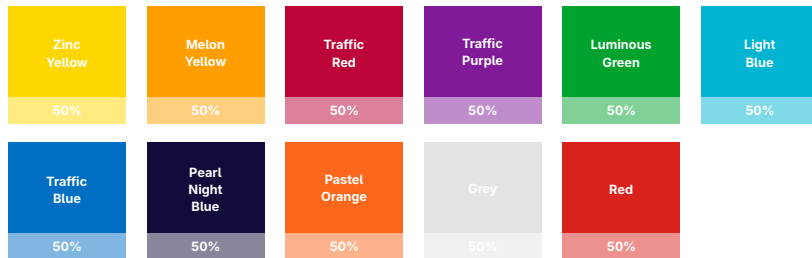


See <https://www.eur.nl/en/about-eur/house-style/brand-elements/colours>.

Attention!

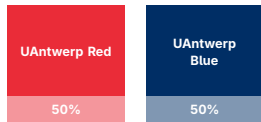
Erasmus University Color Palette (2)

Secondary color palette



University of Antwerp Color Palette (1)

Main colors



See <https://www.uantwerpen.be/en/projects/uantwerp-style-guide/colours/>.

Attention!

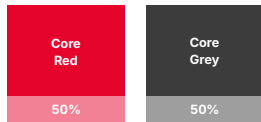
University of Antwerp Color Palette (2)

Faculty colors



University of Leicester Color Palette (1)

Core colors

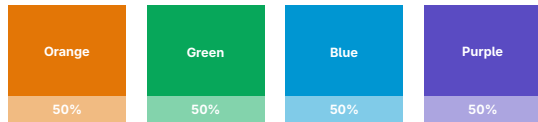


See <https://le.ac.uk/-/media/uol/docs/recruitment/brand-guidelines/uol-quick-guidelines-2020.pdf>.

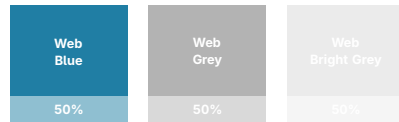
Attention!

University of Leicester Color Palette (2)

Core palette



Web colors

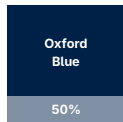


Supporting palette



University of Oxford Color Palette (1)

Primary color



See <https://communications.web.ox.ac.uk/communications-resources/visual-identity/identity-guidelines/colours>.

Attention!

University of Oxford Color Palette (2)

Secondary colors



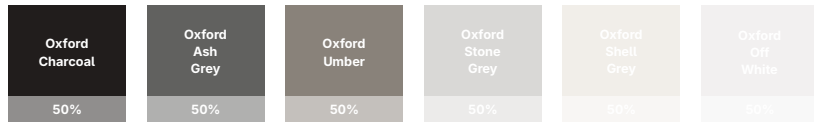
University of Oxford Color Palette (3)

Secondary colors (continued)

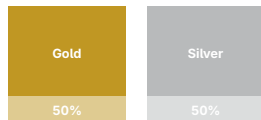


University of Oxford Color Palette (4)

Neutrals

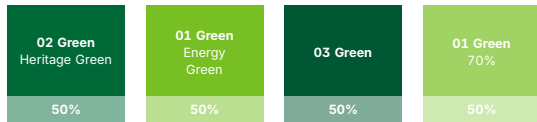


Metallic colors



University of Stirling Color Palette (1)

Primary colors



See <https://www.stir.ac.uk/brand-bank/brand-assets/colour-palette/>.

Attention!

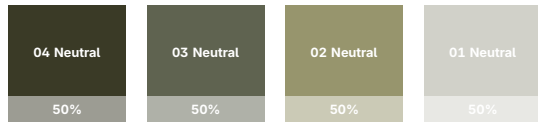
University of Stirling Color Palette (2)

Secondary colors

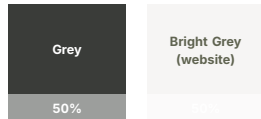


University of Stirling Color Palette (3)

Neutral colors



Tertiary colors



University of Stirling Color Palette (4)

Text samples

03 Green

03 Teal

03 Blue

03 Purple

03 Pink

03 Orange

03 Yellow

03 Neutral

04 Neutral

02 Green/Heritage Green

02 Teal

02 Blue

02 Purple

02 Pink

02 Orange

02 Yellow

02 Neutral

00 Grey

01 Green/Energy Green

01 Teal

01 Blue

01 Purple

01 Pink

01 Orange

01 Yellow

01 Neutral

University of Melbourne Color Palette

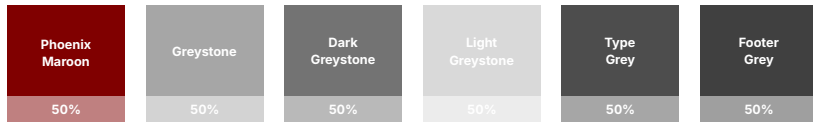


See <https://designsystem.web.unimelb.edu.au/style-guide/colour-palette/>.

Attention!

University of Chicago Color Palette (1)

Primary digital color palette



See <https://uchicago.app.box.com/s/fnrmbbycoxhxt4z2ae46ljvo3urmbvlqq>.

Attention!

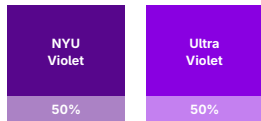
University of Chicago Color Palette (2)

Secondary palette

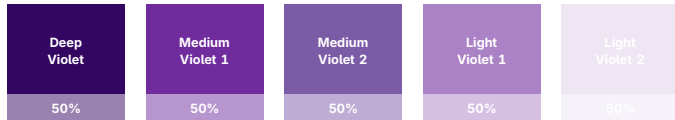


New York University Color Palette (1)

Primary colors



Secondary colors

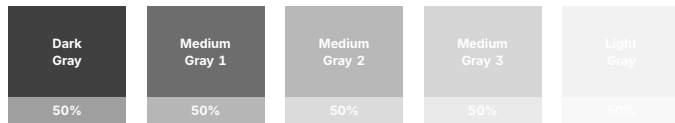


See <https://www.nyu.edu/employees/resources-and-services/media-and-communications/nyu-brand-guidelines/designing-in-our-style/nyu-colors.html>.

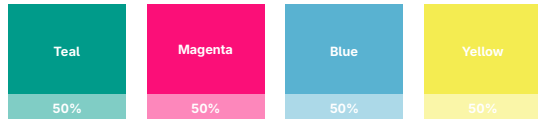
Attention!

New York University Color Palette (2)

Neutral colors

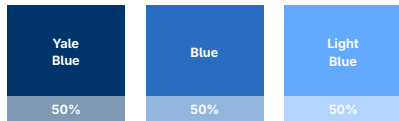


Accent colors

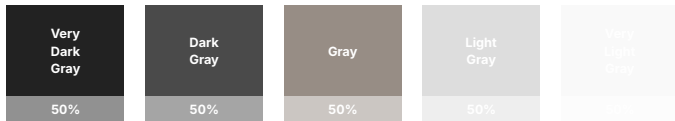


Yale University Color Palette

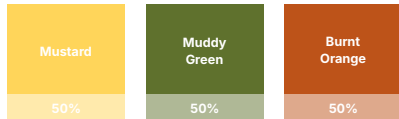
Blues



Grays



Accent colors



See <https://yaleidentity.yale.edu/guidelines/websites>.

Attention!

6

Results

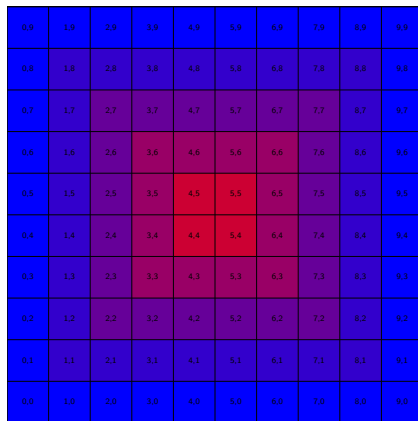


Figure 2. An impressive graph

Results: An siunitx example table

Table 1. Example of a regression table

	(1)	(2)	(3)	(4)	(5)
Treatment	−0.390 (+0.352)	−0.228 (−0.205)	−0.729* [+0.377]	−0.449* [−0.245]	−0.453** {+0.204}
Female	0.948*** (0.354)	0.061 (0.233)	0.188 (0.372)	0.305 (0.226)	0.385* (0.222)
Female × Treatment	0.169 (0.514)	0.251 (0.325)	0.892* (0.533)	0.454 (0.341)	0.439 (0.307)
Final high school grade	−0.101 (0.198)	0.013 (0.144)	0.076 (0.224)	0.117 (0.146)	0.039 (0.133)
Trait self-control	−0.016 (0.016)	0.002 (0.010)	−0.016 (0.015)	0.000 (0.010)	−0.007 (0.009)
Constant	2.357*** (0.239)	1.512*** (0.144)	−0.322 (0.265)	2.158*** (0.161)	1.437*** (0.152)
Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment × (1 + Female)	−0.221	0.023	0.163	0.004	−0.014
p_F [Treatment × (1 + Female) = 0]	0.327	0.008	0.192	0.000	0.003

Dependent variable: $m_{\sim}$. Robust standard errors (cluster-corrected for column 5) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Missing observations ($N < 308$) due to exclusion of trials in which subjects behaved irrationally (i.e., chose a dominated option). The regressors Final high school grade and Trait self-control are mean-centered.

Results: An siunitx example table

Table 2. Example of a regression table

	(1)	(2)	(3)	(4)	(5)
Treatment	−0.390 (+0.352)	−0.228 (−0.205)	−0.729* [+0.377]	−0.449* [−0.245]	−0.453** {+0.204}
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Female × Treatment	0.169 (0.514)	0.251 (0.325)	0.892* (0.533)	0.454 (0.341)	0.439 (0.307)
Final high school grade	−0.101 (0.198)	0.013 (0.144)	0.076 (0.224)	0.117 (0.146)	0.039 (0.133)
Trait self-control	−0.016 (0.016)	0.002 (0.010)	−0.016 (0.015)	0.000 (0.010)	−0.007 (0.009)
Constant	2.357*** (0.239)	1.512*** (0.144)	−0.322 (0.265)	2.158*** (0.161)	1.437*** (0.152)
Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment × (1 + Female)	−0.221	0.023	0.163	0.004	−0.014
$p_F[\text{Treatment} \times (1 + \text{Female}) = 0]$	0.327	0.008	0.192	0.000	0.003

Dependent variable: m_{it} . Robust standard errors (cluster-corrected for column 5) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Missing observations ($N < 308$) due to exclusion of trials in which subjects behaved irrationally (i.e., chose a dominated option). The regressors Final high school grade and Trait self-control are mean-centered.

7

Conclusion

Conclusion

- We have terrific data.
- We have highly convincing findings.
- No-one has done this before
 - Our unique dataset made this possible.
 - Similar attempts: Hostetler et al. (1998), Sigfridsson and Ryde (1998), Markey (2005), Padhye et al. (1999)
- **We are so happy.**

Thank You!

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<https://www.econ.uni-bonn.de/iame/gerhardt>



Math "Torture" Test

Math “Torture” Test: Latin and Greek glyphs (1/3)

$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \varsigma, \tau, \upsilon, \varphi, \chi, \psi, \omega, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, -g, \gamma, \gamma, \Gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \varphi, \varphi, \chi, \psi, \omega, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, -g, \gamma, \gamma, \Gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \varphi, \varphi, \chi, \psi, \omega, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, -g, \gamma, \gamma, \Gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \varsigma, \tau, \upsilon, \varphi, \chi, \psi, \omega, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, -g, \gamma, \gamma, \Gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \varphi, \varphi, \chi, \psi, \omega, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, -g, \gamma, \gamma, \Gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \varphi, \varphi, \chi, \psi, \omega, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, -g, \gamma, \gamma, \Gamma, \Gamma$

Math “Torture” Test: Latin and Greek glyphs (2/3)

$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \mathbb{F}, \mathbb{A}, \mathbb{B}, \Gamma, \Delta, \mathbb{E}, \mathbb{Z}, \mathbb{H}, \Theta, \mathbb{I}, \mathbb{K}, \mathbb{L}, \mathbb{M}, \mathbb{N}, \Xi, \mathbb{O}, \Pi, \mathbb{P}, \Sigma, \mathbb{T}, \mathbb{Y}, \Phi, \mathbb{X}, \Psi, \Omega, \mathbb{F}, -g, \gamma, \gamma, \Gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \mathbb{F}, \mathbb{A}, \mathbb{B}, \Gamma, \Delta, \mathbb{E}, \mathbb{Z}, \mathbb{H}, \Theta, \mathbb{I}, \mathbb{K}, \mathbb{L}, \mathbb{M}, \mathbb{N}, \Xi, \mathbb{O}, \Pi, \mathbb{P}, \Sigma, \mathbb{T}, \mathbb{Y}, \Phi, \mathbb{X}, \Psi, \Omega, \mathbb{F}, -g, \gamma, \gamma, \Gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \mathbb{F}, \mathbb{A}, \mathbb{B}, \Gamma, \Delta, \mathbb{E}, \mathbb{Z}, \mathbb{H}, \Theta, \mathbb{I}, \mathbb{K}, \mathbb{L}, \mathbb{M}, \mathbb{N}, \Xi, \mathbb{O}, \Pi, \mathbb{P}, \Sigma, \mathbb{T}, \mathbb{Y}, \Phi, \mathbb{X}, \Psi, \Omega, \mathbb{F}, -g, \gamma, \gamma, \Gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \mathbb{F}, \mathbb{A}, \mathbb{B}, \Gamma, \Delta, \mathbb{E}, \mathbb{Z}, \mathbb{H}, \Theta, \mathbb{I}, \mathbb{K}, \mathbb{L}, \mathbb{M}, \mathbb{N}, \Xi, \mathbb{O}, \Pi, \mathbb{P}, \Sigma, \mathbb{T}, \mathbb{Y}, \Phi, \mathbb{X}, \Psi, \Omega, \mathbb{F}, -g, \gamma, \gamma, \Gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \mathbb{F}, \mathbb{A}, \mathbb{B}, \Gamma, \Delta, \mathbb{E}, \mathbb{Z}, \mathbb{H}, \Theta, \mathbb{I}, \mathbb{K}, \mathbb{L}, \mathbb{M}, \mathbb{N}, \Xi, \mathbb{O}, \Pi, \mathbb{P}, \Sigma, \mathbb{T}, \mathbb{Y}, \Phi, \mathbb{X}, \Psi, \Omega, \mathbb{F}, -g, \gamma, \gamma, \Gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \mathbb{F}, \mathbb{A}, \mathbb{B}, \Gamma, \Delta, \mathbb{E}, \mathbb{Z}, \mathbb{H}, \Theta, \mathbb{I}, \mathbb{K}, \mathbb{L}, \mathbb{M}, \mathbb{N}, \Xi, \mathbb{O}, \Pi, \mathbb{P}, \Sigma, \mathbb{T}, \mathbb{Y}, \Phi, \mathbb{X}, \Psi, \Omega, \mathbb{F}, -g, \gamma, \gamma, \Gamma, \Gamma$

Math “Torture” Test: Latin and Greek glyphs (3/3)

Disambiguation: 0 O O, 1 l l | l l /, i j, r n m, ε , θ Θ , ϕ ψ , ω , — —

Latin vs. Greek: a α , d δ , e ε , i ι , k κ , n η , o σ , p ρ , β β , u v , v v , w ω , x χ , y γ , A Δ Λ , O Θ Ω , T Γ , Y Y .

Disambiguation: 0 O O, 1 l l | l l /, i j, r n m, θ Θ , ϕ ψ , — —

Latin vs. Greek: a α , d δ , e ε , i ι , k κ , n η , o σ , p ρ , β β , u u , v v , w ω , x χ , y γ , A Δ Λ , O Θ Ω , T Γ , Y Y .

Theorem (Residue Theorem)

Let f be analytic in the region G except for the isolated singularities $a_1, a_2, \dots, a_m$. If γ is a closed rectifiable curve in G which does not pass through any of the points a_k and if $\gamma \approx 0$ in G , then

$$\frac{1}{2\pi i} \int_{\gamma} f = \sum_{k=1}^m n(\gamma; a_k) \operatorname{Res}(f; a_k).$$

Theorem (Maximum Modulus)

Let G be a bounded open set in $\mathbb{C}$ and suppose that f is a continuous function on G^- which is analytic in G . Then

$$\max\{|f(z)| : z \in G^-\} = \max\{|f(z)| : z \in \partial G\}.$$

Math “Torture” Test: Various expressions and formulas (3/20)

Sample page of mathematical typesetting

First some large operators, both in text: $\int\int\int\limits_{\mathcal{Q}} f(x,y,z) \, dx \, dy \, dz$ and $\prod_{Y \in \Gamma_{\tilde{C}}} \partial(\tilde{X}_Y)$; and also on display:

$$\int\int\int\int\limits_{\mathbf{Q}} f(w,x,y,z) \, dw \, dx \, dy \, dz \leq \oint_{\partial \mathbf{Q}} f' \left(\max \left\{ \frac{\|w\|}{|w^2 + x^2|}, \frac{\|z\|}{|y^2 + z^2|}, \frac{\|w \oplus z\|}{\|x \oplus y\|} \right\} \right)$$
$$\approx \bigcup_{Q \in \bar{\mathbf{Q}}} \left[f^* \left(\frac{\int Q(t)}{\sqrt{1-t^2}} \right) \right]_{t=\alpha}^{t=}$$
(1)

Math “Torture” Test: Various expressions and formulas (4/20)

For x in the open interval $] -1, 1[$ the infinite sum in Equation (2) is convergent; however, this does not hold throughout the closed interval $[-1, 1]$.

$$(1 - x)^{-k} = 1 + \sum_{j=1}^{\infty} (-1)^j \binom{k}{j} x^j \quad \text{for } k \in \mathbb{N}; k \neq 0. \quad (2)$$

Math “Torture” Test: Various expressions and formulas (5/20)

Most of the following examples are taken from *The T_EXbook* (Knuth 1984, see <https://ctan.org/pkg/texbook>) and were adapted for L^AT_EX from Karl Berry’s torture test for plain T_EX math fonts.

Math “Torture” Test: Various expressions and formulas (6/20)

$x + y - z, \quad x + y * z, \quad z * y / z, \quad (x + y)(x - y) = x^2 - y^2,$
 $x \times y \cdot z = [x y z], \quad x \circ y \bullet z, \quad x \cup y \cap z, \quad x \sqcup y \sqcap z,$
 $x \vee y \wedge z, \quad x \pm y \mp z, \quad x = y / z, \quad x := y, \quad x \leq y \neq z, \quad x \sim y \simeq z \quad x \equiv y \not\equiv z, \quad x \subset y \subseteq z$
 $\sin 2\theta = 2 \sin \theta \cos \theta, \quad O(n \log n \log n), \quad \Pr(X > x) = \exp(-x/\mu),$
 $(x \in A(n) \mid x \in B(n)), \quad \bigcup_n X_n \parallel \bigcap_n Y_n$

$x + y - z, \quad x + y * z, \quad z * y / z, \quad (x + y)(x - y) = x^2 - y^2,$
 $x \times y \cdot z = [x y z], \quad x \circ y \bullet z, \quad x \cup y \cap z, \quad x \sqcup y \sqcap z,$
 $x \vee y \wedge z, \quad x \pm y \mp z, \quad x = y / z, \quad x := y, \quad x \leq y \neq z, \quad x \sim y \simeq z \quad x \equiv y \not\equiv z, \quad x \subset y \subseteq z$
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 $(x \in A(n) \mid x \in B(n)), \quad \bigcup_n X_n \parallel \bigcap_n Y_n$

Math “Torture” Test: Various expressions and formulas (7/20)

In-text matrices $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} a & b & c \\ 1 & m & n \end{pmatrix}$. \$ % ! ? & . , ; :

In-text matrices $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} a & b & c \\ 1 & m & n \end{pmatrix}$. \$ % ! ? & . , ; :

Math “Torture” Test: Various expressions and formulas (8/20)

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4}}}}$$

$$\binom{p}{2} x^2 y^{p-2} - \frac{1}{1-x} \frac{1}{1-x^2} = \frac{a+1}{b} \Bigg/ \frac{c+1}{d}.$$

Math “Torture” Test: Various expressions and formulas (9/20)

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4}}}}$$

$$\binom{p}{2} x^2 y^{p-2} - \frac{1}{1-x} \frac{1}{1-x^2} = \frac{a+1}{b} \Bigg/ \frac{c+1}{d}.$$

Math “Torture” Test: Various expressions and formulas (10/20)

$$\sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + x}}}}}$$
$$\sqrt[n]{1 + \sqrt[k]{1 + \sqrt[5]{1 + \sqrt[4]{1 + \sqrt[3]{1 + x}}}}}$$

Math “Torture” Test: Various expressions and formulas (11/20)

$$\sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + x}}}}}$$
$$\sqrt[n]{1 + \sqrt[k]{1 + \sqrt[5]{1 + \sqrt[4]{1 + \sqrt[3]{1 + x}}}}}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |(x + iy)|^2 = 0$$

$$\pi(n) = \sum_{m=2}^n \left[\left(\sum_{k=1}^{m-1} \lfloor (m/k) \rfloor / \lceil m/k \rceil \right)^{-1} \right].$$

$$\int_0^\infty \frac{t - ib}{t^2 + b^2} e^{iat} dt = e^{ab} E_1(ab), \quad a, b > 0.$$

$$\mathbf{A} := \begin{bmatrix} x - \lambda & 1 & 0 \\ 0 & x - \lambda & 1 \\ 0 & 0 & x - \lambda \end{bmatrix}.$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |(x+iy)|^2 = 0$$

$$\pi(n) = \sum_{m=2}^n \left\lfloor \left(\sum_{k=1}^{m-1} \lfloor (m/k) \rfloor / \lceil m/k \rceil \right)^{-1} \right\rfloor.$$

$$\int_0^\infty \frac{t-ib}{t^2+b^2} e^{iat} dt = e^{ab} E_1(ab), \quad a, b > 0.$$

$$A := \begin{bmatrix} x-\lambda & 1 & 0 \\ 0 & x-\lambda & 1 \\ 0 & 0 & x-\lambda \end{bmatrix}.$$

Math "Torture" Test: Various expressions and formulas (14/20)

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \begin{pmatrix} u & x \\ v & y \\ w & z \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

$$\mathbf{M} = \begin{matrix} & C & I & C' \\ \begin{matrix} C \\ I \\ C' \end{matrix} & \begin{pmatrix} 1 & 0 & 0 \\ b & 1-b & 0 \\ 0 & a & 1-a \end{pmatrix} \end{matrix}$$

$$\sum_{n=0}^{\infty} a_n z^n \text{ converges if } |z| < \left(\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \right)^{-1}.$$

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \rightarrow f'(x) \quad \text{as } \Delta x \rightarrow 0.$$

$$\|u_i\| = 1, \quad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

The confluent image of $\left\{ \begin{array}{l} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{array} \right\}$ is $\left\{ \begin{array}{l} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{array} \right\}.$

$$\sum_{n=0}^{\infty} a_n z^n \quad \text{converges if} \quad |z| < \left(\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \right)^{-1}.$$

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \rightarrow f'(x) \quad \text{as } \Delta x \rightarrow 0.$$

$$\|u_i\| = 1, \quad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of } \left\{ \begin{array}{l} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{array} \right\} \quad \text{is} \quad \left\{ \begin{array}{l} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{array} \right\}.$$

$$\begin{aligned}T(n) &\leq T(2^{\lceil \lg n \rceil}) \leq c(3^{\lceil \lg n \rceil} - 2^{\lceil \lg n \rceil}) \\&< 3c \cdot 3^{\lg n} \\&= 3c n^{\lg 3}.\end{aligned}$$

$$\begin{aligned}(x + y)(x - y) &= x^2 - xy + yx - y^2 \\&= x^2 - y^2 \\(x + y)^2 &= x^2 + 2xy + y^2.\end{aligned}$$

$$\begin{aligned}\left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} dr d\theta \\ &= \int_0^{2\pi} \left(e^{-\frac{r^2}{2}} \Big|_{r=0}^{r=\infty} \right) d\theta \\ &= \pi.\end{aligned}$$

$$\prod_{k \geq 0} \frac{1}{(1 - q^k z)} = \sum_{n \geq 0} z^n \bigg/ \prod_{1 \leq k \leq n} (1 - q^k).$$

$$\sum_{\substack{0 < i \leq m \\ 0 < j \leq n}} p(i, j) \neq \sum_{i=1}^p \sum_{j=1}^q \sum_{k=1}^r a_{ij} b_{jk} c_{ki} \neq \sum_{\substack{1 \leq i \leq p \\ 1 \leq j \leq q \\ 1 \leq k \leq r}} a_{ij} b_{jk} c_{ki}$$

Math “Torture” Test: Various expressions and formulas (20/20)

$$\max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Inline math:

$$\max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$p_1(n) = \lim_{m \rightarrow \infty} \sum_{v=0}^{\infty} (1 - \cos^{2m}(v!^n \pi/n))$$

Inline math:

$$p_1(n) = \lim_{m \rightarrow \infty} \sum_{v=0}^{\infty} (1 - \cos^{2m}(v!^n \pi/n))$$

9

References

References (1/2)

- Aristotle. 1907. *De Anima*. Edited by Robert Drew Hicks. Cambridge: Cambridge University Press.
- Aristotle. 1929. *Physics*. Translated by P. H. Wicksteed and F. M. Cornford. New York: G. P. Putnam.
- Aristotle. 1968. *Poetics*. Edited by D. W. Lucas. Clarendon Aristotle. Oxford: Clarendon Press.
- Chiu, Willy W., and We Min Chow. 1978. "A Hybrid Hierarchical Model of a Multiple Virtual Storage (MVS) Operating System." Research report RC-6947. IBM.
- Gerhardt, Michael J. 2000. *The Federal Appointments Process: A Constitutional and Historical Analysis*. Durham and London: Duke University Press.
- Goossens, Michel, Frank Mittelbach, and Alexander Samarin. 1994. *The LaTeX Companion*. 1st ed. Reading, Mass.: Addison-Wesley.
- Hostetler, Michael J., Julia E. Wingate, Chuan-Jian Zhong, Jay E. Harris, Richard W. Vachet, Michael R. Clark, J. David Londono, et al. 1998. "Alkanethiolate gold cluster molecules with core diameters from 1.5 to 5.2 nm: Core and monolayer properties as a function of core size." *Langmuir* 14 (1): 17–30.
- Knuth, Donald E. 1984. *The TeXbook*. Vol. A of *Computers & Typesetting*. Reading, Mass.: Addison-Wesley.
- Markey, Nicolas. 2005. "Tame the BeaST: The B to X of BibTeX," October 16, 2005. Accessed October 1, 2006. http://mirror.ctan.org/info/bibtex/tamethebeast/ttb_en.pdf.
- Padhye, Jitendra, Victor Firoiu, and Don Towsley. 1999. "A Stochastic Model of TCP Reno Congestion Avoidance and Control." Technical report 99-02. Amherst, Mass.: University of Massachusetts.

References (2/2)

- Sarfraz, M., and M. F. A. Razzak. 2002. "Technical section: An algorithm for automatic capturing of the font outlines." *Computers and Graphics* 26 (5): 795–804.
- Sigfridsson, Emma, and Ulf Ryde. 1998. "Comparison of methods for deriving atomic charges from the electrostatic potential and moments." *Journal of Computational Chemistry* 19 (4): 377–395. [https://doi.org/10.1002/\(SICI\)1096-987X\(199803\)19:4<377::AID-JCC1>3.0.CO;2-P](https://doi.org/10.1002/(SICI)1096-987X(199803)19:4<377::AID-JCC1>3.0.CO;2-P).
- Van Gennep, Arnold. 1909. *Les rites de passage*. Paris: Nourry.

**Thank you for your attention—
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